

Predicting Airline Flight Arrival Delays Based on Weather Conditions

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1 Introduction

Flight delays are a persistent challenge in aviation logistics, often driven by scheduling, market dynamics, and environmental factors. While individual weather metrics (temperature, wind speed, precipitation) are known to affect operations, these variables are highly collinear. For example, high wind speed and rain often occur together.

A flight delay has a great impact on airline companies as they greatly decrease customer satisfaction as evidenced in Song et al.'s 2024 study. They found that flight delays not only cost airlines but also negatively impact passengers' travel plans and experiences. There are also more indirect effects on a country's economy as they hinder the development of sectors that support the air transport industry (Chen et al., 2018).

Current research often treats these variables as independent features, which can lead to overfitting or models that fail to capture the holistic "severity" of a weather event. In Choi et al.'s 2016 paper on predicting weather-induced airline delays, the team used supervised machine learning methods including decision trees, random forest, and ADA boost. It also focused solely on flights departing from Denver International Airport and arriving in Charlotte Douglas International Airport which fails to take into account regional differences in handling of weather delays. Similar to how public school in the north are able to better handle snow delays it is likely airports in regions that experience common weather states are better equipped to handle them.

If we can successfully discover how localized weather states interact with these geographic and scheduling variables within the data, the aviation industry's approach to delay management will fundamentally change. Instead of relying on reactive models that simply expect a delay when a storm hits a specific airport, airlines could develop proactive, network-wide predictive models. Discovering the exact mathematical weight of geographic routing versus weather severity would allow carriers to anticipate delays hours in advance, allowing them to reroute physical planes to avoid trapping them in regions unequipped for specific weather states.

Ethically and logistically, utilizing historical flight data requires acknowledging that past weather patterns may not perfectly mirror future occurrences due to climate shifts. New

technologies have also improved the aircraft’s abilities to endure certain weather conditions so something that caused a delay 15 years ago might not cause a delay today. The raw data utilized for this research is the Airline On-Time Performance Data (OTPD) dataset, collected by the US Bureau of Transportation Statistics, the FAA, and the National Center for Environmental Information. It is freely available via Mendeley (<https://data.mendeley.com/datasets/j3z5bm7496/1>).

2 Data Explanation or Exploration

The OTPD dataset is comprehensive, containing over a million instances of US domestic flights recorded between 2004 and 2017. A significant advantage of this dataset is its geographic diversity, which captures flights across the entire country, enabling an analysis of how regional airports handle varying weather severities.

My features of interest were initially isolated to 16 environmental variables, including continuous features like windspeed and temperature, alongside categorical dummy variables for rain, snow, and temperature bins. The target variable was `arrdelay` (arrival delay). The dataset had relatively few missing values relative to the size of the dataset, and rows missing data were usually missing it across most variables; therefore, I opted to drop rows with any missing values rather than imputing them.

For the predictive validation phase, the data was split chronologically, using 2004–2014 as the training set and 2015–2017 as the testing set. This mimics the real-world application of forecasting future delays from past historical trends.

Initially, the primary response variable was `arrdelay` (arrival delay), which is a continuous numeric variable representing the exact number of minutes a flight was early or late. However, exploratory data analysis revealed that this continuous variable is incredibly skewed; the vast majority of flights arrive early or on time, while a small minority experience massive, multi-hour delays. Because predicting the exact minute of a delay is exceptionally difficult and less operationally useful than identifying severe disruptions, I transformed the continuous `arrdelay` variable into a binned categorical variable.

Using four distinct bins, the flights were classified as "Early" (< 0 minutes), "On-time or Small Delay" (0 to 14 minutes), "Moderate Delay" (15 to 44 minutes), and "Major Delay" (45+ minutes). This transformation allowed the models to shift from a regression task to a classification task.

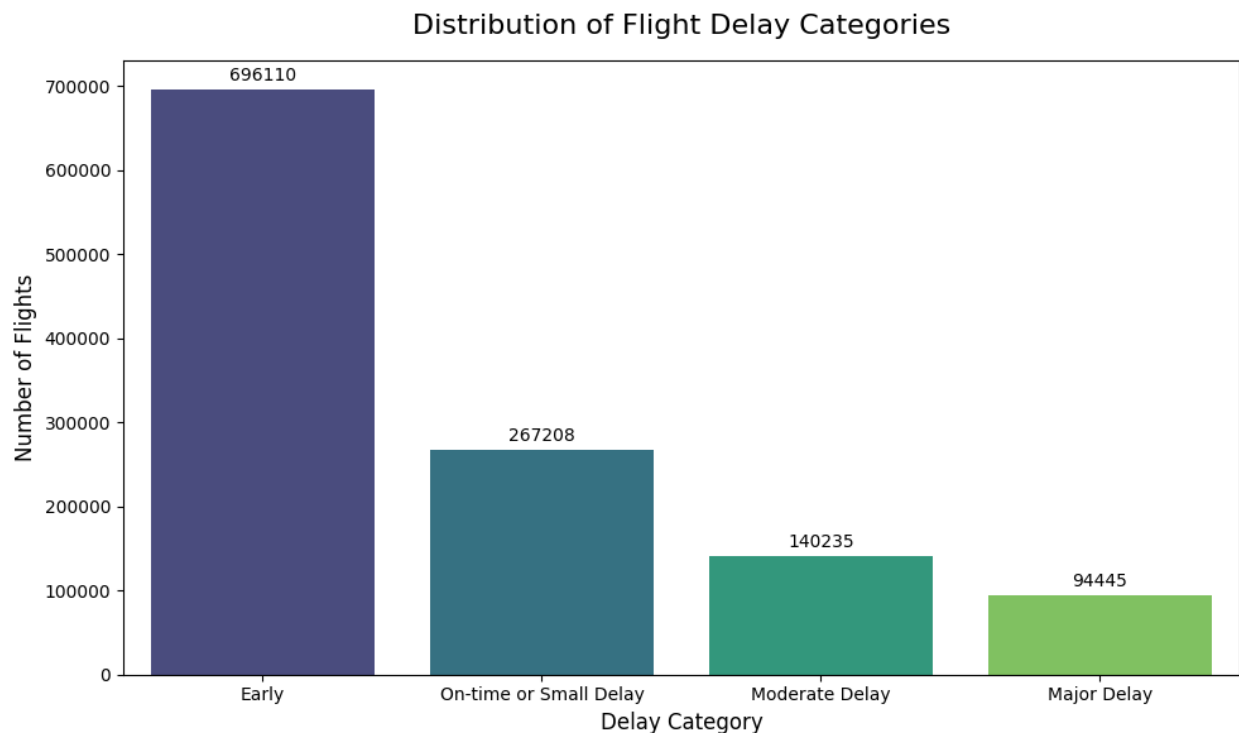


Figure 1: Count of Flights in Each Delay Category

3 Algorithms and Techniques Analysis

Because finding the best way to model the interaction between weather and delays is complex, this project underwent four distinct algorithmic iterations.

3.1 Iteration 1: Principal Component Analysis (PCA) and Linear Regression

Initially, I hypothesized that an unsupervised machine learning technique, PCA, would be the most effective strategy. Often weather conditions like rain, temperature, and wind are highly correlated. By compressing these raw sensor measurements into a compact set of linearly uncorrelated principal components (or "weather states"), I hoped to prevent downstream regression models from overfitting and be able to predict delays based on weather events like snowstorms, windy days, etc. I thought a weather state would be easier to interpret as airlines could understand the impacts of weather events rather than raw numbers. I applied a `StandardScaler()` prior to fitting the PCA so continuous variables would not dominate the binary dummy variables.

The PCA failed to condense the variables efficiently. The first three principal components explained only 43.04% of the total variance, and it required 9 components to reach my 80% threshold. Furthermore, when I fed these components into a Linear Regression model to predict `arrdelay`, the model yielded an R^2 of 0.0018—explaining effectively 0% of the variance in delay times. I did more research into PCA and found that it struggles with binary dummy

variables (which comprised of 12 of my 16 variables), and Linear Regression cannot capture the non-linear, interactive nature of weather (e.g., 10mph winds on a sunny day vs. during a snowstorm).

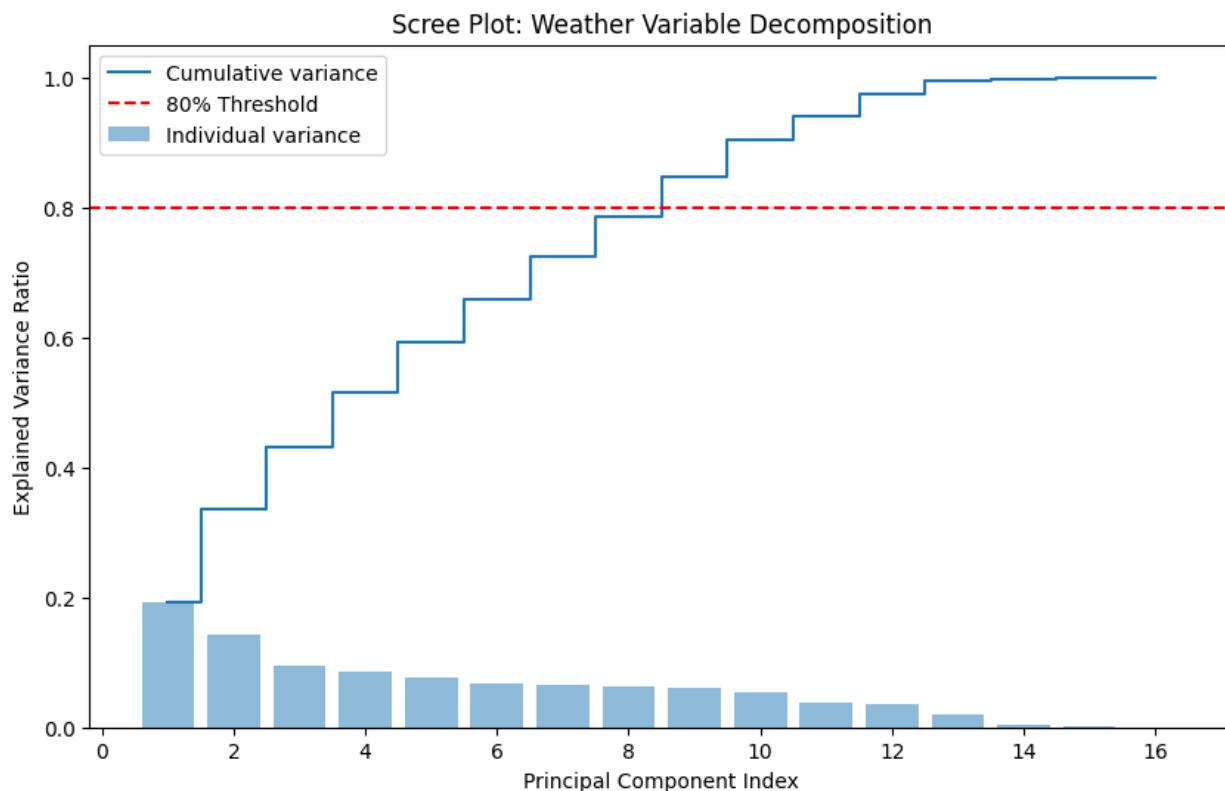


Figure 2: Scree Plot for PCA

3.2 Iteration 2 & 3: Random Forest Regression & Classification (Weather Only)

To address non-linearity and dummy variables, I pivoted to a tree-based model. I first ran a Random Forest Regressor on the raw, unscaled weather variables. While the results improved slightly over linear regression, predicting the exact minute of a flight delay proved to be difficult as shown in my performance metrics.

I then reframed the problem into a classification task using a RandomForestClassifier. I binned the arrival delays into four categories: "Early" (<10 mins), "On-time or Small Delay" (0-15 mins), "Moderate Delay" (15-45 mins), and "Major Delay" (45+ mins). I utilized `class-weight='balanced'` because the vast majority of flights arrive early or on time, and I wanted to penalize the model for missing rare Major Delays which are detrimental to airlines.

The model achieved an overall accuracy of 43.32% and successfully caught 26% of all Major Delays (Recall: 0.26). However, the confusion matrix revealed massive false positives (predicting bad weather would cause a delay, but the plane arrived early) and false negatives (predicting good weather meant an early arrival, but the plane was over 45 minutes late). This proved weather alone could not explain the delays.

3.3 Iteration 4: Random Forest Classification (Weather + Geography/Logistics)

To align with my initial proposal regarding regional differences, I added geographic and logistical variables to the Random Forest model: origin, dest, uniquecarrier, and scheduledhour. These variables described where the plane was arriving and departing, the airline company, and the time the flight was scheduled for. I hypothesized that the scheduledhour variable would be important in predicting delay due to cascading delays (if a morning flight is delayed, the later flights using that same plane will probably be delayed). Machine learning models require numeric inputs, so I used an OrdinalEncoder to transform the categorical text variables (like airport codes) into integers before training.

While overall accuracy shifted slightly to 41.44%, the model's ability to catch severe disruptions skyrocketed. The recall for Major Delays nearly doubled to 46% (0.46).

3.4 Comparing Iterations

Iteration	Model Type	Features	Target Task	R ²	Accuracy	Recall
Iteration 1	Linear Regression	Weather (PCA)	Regression (Minutes)	0.0018	N/A	N/A
Iteration 2	Random Forest Regressor	Weather (Raw)	Regression (Minutes)	0.0180	N/A	N/A
Iteration 3	Random Forest Classifier	Weather (Raw)	Classification (Bins)	N/A	43.32%	0.26
Iteration 4	Random Forest Classifier	Weather + Geography/Logistics	Classification (Bins)	N/A	41.44%	0.46

4 Conclusion

Ultimately, the iterative process of this project showed that weather acts as a catalyst for delays, but geography and scheduling also plays a large part in the severity of a delay.

When observing the final Feature Importances chart from the Random Forest model, the time of day (scheduledhour), the destination (dest), and the origin (origin) ranked as the top three most important predictors of flight delays, easily outpacing direct weather events like windspeed or snowdummy. The model proved that an early morning flight out of a major hub in a snowstorm is often less likely to be delayed than a late-night flight on a perfectly sunny day that is suffering from the cascading network effects of a delayed plane from earlier in the afternoon.

While this is an interesting finding, my goal of this project was to investigate how weather conditions affected delays. A major limitation of this study was that the dataset did not give any reason for each flight's delay. Future analysis could be improved by using a dataset that gives a reason for each flight delay to filter out flights that were delayed because of previous cascading delays or mechanical issues.

This work does raise the question of how cascading delays can be dealt with. Could better prediction of weather delays be key in minimizing the effects of one delay on subsequent flights? Future research isolated on weather delays could help inform aviation strategy moving forward when it comes to these cascading delays.

References

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